Parametric & Sequences Test, Day 2

Cram sheet

Friday, calculator: the nth term, finding sums, the Binomial Theorem + 1 projectile. 27 points.

Memorize: not given on the test

(1) $\sum ca_k = c\sum a_k$

(2), (3) $\sum(a_k \pm b_k) = \sum a_k \pm \sum b_k$

(4) $\displaystyle\sum_{k=j+1}^{n} a_k = \sum_{k=1}^{n} a_k - \sum_{k=1}^{j} a_k$ (j = start -1)

$0! = 1,\ 1! = 1$

(5) $\sum_{k=1}^{n} 1 = n$ (6) $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$

(7) $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$

(8) $\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2$

Find a sum

1. Lower index not 1: **fix it with (4) first**. Start at $k = 6$ means $j = 5$.
2. Split with (2), (3); constants out with (1).
3. Write each formula from memory with n plugged in. **This line is required before the calculator.** The Σ button is blocked in test mode.

Ex: $\sum_{k=6}^{20} 4k^2 = 4\left[\frac{20(21)(41)}{6} - \frac{5(6)(11)}{6}\right] = 11260$ $\sum_{k=1}^{10}(k^3+1) = \left[\frac{10(11)}{2}\right]^2 + 10 = 3035$

Binomial Theorem: Pascal's triangle

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1
1 8 28 56 70 56 28 8 1

- Outsides 1; inside = sum of the two above. Row n for power n (up to about 7 or 8).
- Blanks first: first term's powers **down** $n \to 0$, second term's **up** $0 \to n$. Fill with row n, simplify.
- The sign goes with its term; signs alternate only when the second term is negative. Use Pascal's triangle: multiplying it out loses credit.
- $(2x)^4 = 16x^4$, not $2x^4$.

Ex: $(2x-3)^4 = 16x^4 - 96x^3 + 216x^2 - 216x + 81$

The nth term (runs a little harder than the homework)

Write 1 2 3 4 above the terms; find the top and the bottom separately; first term positive $(-1)^{n+1}$, negative $(-1)^n$; odds $2n-1$, evens $2n$, squares n^2, powers b^{n-1}. Check $n = 1$.

Ex: $\frac{1}{3}, -\frac{2}{9}, \frac{3}{27}, -\frac{4}{81}, \ldots \to a_n = (-1)^{n+1}\frac{n}{3^n}$ $2, -4, 6, -8, \ldots \to a_n = 2n(-1)^{n+1}$

Σ notation

- **Write out a sum:** start at the lower index (it can be $k = 0$), first few terms, ellipsis, last term with n; don't add.
- **Express as Σ:** lower index, upper index, general term, same letter. Check the first and last terms.

Ex: $\frac{1}{3} + \frac{2}{4} + \cdots + \frac{15}{17} = \sum_{k=1}^{15} \frac{k}{k+2}$ $\sum_{k=1}^{n}(k^2+1) = 2 + 5 + 10 + \cdots + (n^2+1)$

Projectile

1. Equations with the numbers and the **degree symbol** (leaving it off is not okay); from the ground $h = 0$.
2. Degree mode, Graph, Parametric, $x_1(t)$, $y_1(t)$ exactly; write the setup. Path doesn't come down? Raise $t$max.
3. Max height: **trace** to the top, read y (Analyze, Maximum doesn't work on parametric). Time: $y \approx 0$, read t. Distance: x at that point. Trace step 0.01, then 0.001.
4. Thousandths, units in feet; "$y \approx$", "$t \approx$", "$x \approx$" are full answers: $y \approx 26.258$ ft, $t \approx 2.512$ s, $x \approx 126.707$ ft.

Where points get lost

- Using a sum formula before fixing a lower index that isn't 1; skipping the substituted-formula line.
- $(2x)^n$ written as $2x^n$; the minus sign left off the second term.
- Leaving off the degree symbol; answers in feet, not meters. Wrong sign exponent on an alternating nth term: check a_1.